

Spiking Neural Networks for Environmental Sound Classification: From Seven Encodings to SpiNNaker Deployment

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ABSTRACT

We present the first convolutional spiking neural network (SNN) evaluation on ESC-50, comparing seven spike encodings and deploying on SpiNNaker neuromorphic hardware. Our SNN achieves 47.15% with direct encoding—a 16.7 pp gap versus a matched ANN (63.85%, $p=0.001$). This gap collapses to 0.95 pp with pretrained features ($p=0.034$), showing the bottleneck is feature learning, not spiking computation. SNNs exhibit $6.0\times$ greater adversarial robustness under FGSM ($\epsilon=0.1$, $p=0.007$). A novel cross-encoding transfer analysis reveals high encoding specificity (transfer ratio 0.255 ± 0.006 , 75% accuracy loss when swapped). We deploy on SpiNNaker via an FC2-only hybrid, achieving $33.1\%\pm6.9\%$ across five folds—the first neuromorphic hardware deployment for environmental sound classification. Code: <https://github.com/ayushkumarcodes/snn-thesis>

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; *Bio-inspired approaches*; • **Hardware** → **Neural systems**.

KEYWORDS

spiking neural networks, environmental sound classification, neuromorphic computing, SpiNNaker, spike encoding

1 INTRODUCTION

The proliferation of always-on audio sensing in edge devices demands both high accuracy and energy efficiency. Convolutional neural networks achieve near-human accuracy on environmental sound benchmarks [1], but their multiply-accumulate (MAC) operations prohibit deployment on battery-constrained devices. Spiking neural networks (SNNs), which communicate via binary spike events, offer an alternative: on neuromorphic hardware such as SpiNNaker [2] and Loihi [19], SNNs execute as accumulate-only operations (ACs) rather than MACs, with per-operation energy $5.1\times$ lower [15].

Despite extensive SNN research on image classification [3] and growing interest in neuromorphic computing [20], **no prior work has evaluated convolutional SNNs on ESC-50**. The closest work [7] evaluates only ESC-10 (10 classes) with fully-connected layers. Yarga et al. [21] compare four spike encodings for speech digits at ICONS 2022; we extend to seven encodings on the harder 50-class ESC-50 task. Dominguez-Morales et al. [8] deploy SNNs on SpiNNaker for audio but evaluate only pure tones, not complex soundscapes.

This paper makes the following contributions:

- (1) **First convolutional SNN on ESC-50** with systematic comparison of seven spike encoding methods, including a novel

cross-encoding transfer analysis revealing high encoding specificity (transfer ratio = 0.255);

- (2) **First SpiNNaker deployment for environmental sound**, using a validated FC2-only hybrid approach as proof of concept, with documented root-cause analysis of full-deployment failure;
- (3) **Gap-collapse finding**: the SNN-ANN gap narrows from 16.7 pp to 0.95 pp with pre-trained features, showing the bottleneck is feature learning, not spiking computation;
- (4) **First adversarial robustness analysis of SNNs on audio**, revealing $6.0\times$ greater robustness under FGSM ($p=0.007$, 5-fold validated).

2 BACKGROUND

ESC-50 [1] contains 2,000 five-second recordings across 50 environmental sound classes organised into five super-categories: Animals, Nature, Human non-speech, Domestic, and Urban. The dataset provides 5 predefined folds for cross-validation. Human performance is 81.3%; ANN state of the art reaches 98–99% using large pre-trained models [14]. We use log-mel spectrograms (64 bins, $n_{\text{fft}}=1024$, hop=512, sr=22050 Hz), yielding 64×216 input tensors. The small training set (1,600 samples per fold split) makes ESC-50 particularly challenging for SNNs, which typically require more data than ANNs to overcome the information loss from binary spike discretisation.

Spiking Neural Networks. Leaky Integrate-and-Fire (LIF) neurons accumulate weighted input, spike when membrane potential exceeds a threshold, and reset:

$$U[t] = \beta \cdot U[t-1] + W \cdot X[t] - S[t-1] \cdot U_{\text{thr}} \quad (1)$$

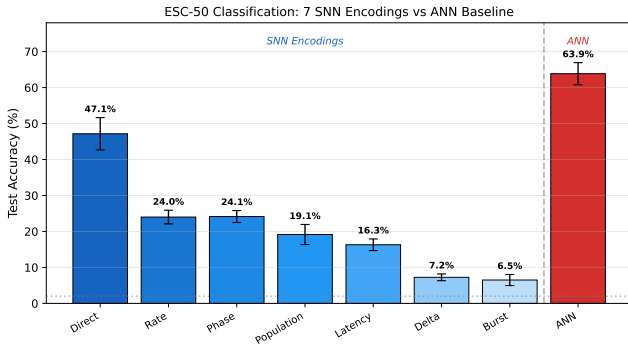
where $U[t]$ is the membrane potential, $\beta=0.95$ is the decay constant, W are synaptic weights, $X[t]$ is the input, and $S[t]$ is the binary output spike ($S[t] = 1$ if $U[t] > U_{\text{thr}}$, else 0). The gradient through the non-differentiable step function $S = \Theta(U - U_{\text{thr}})$ uses a surrogate gradient [4, 5]; we use fast sigmoid (slope=25). Training follows snnTorch [3] with cross-entropy on summed membrane potentials over $T=25$ timesteps.

SpiNNaker [2] is a massively-parallel neuromorphic platform comprising 1 million ARM processors, capable of simulating spiking networks in biological real time. Each core processes up to 256 neurons with integer-quantised weights and spike-driven communication. We use IF_curr_exp neurons with $\tau_{\text{syn}}=5$ ms, $v_{\text{thresh}}=1.0$, $\tau_m=20$ ms, calibrated via a 9-point logarithmic scale sweep.

Prior SNN Audio Work. Larroza et al. [7] classify ESC-10 (10 classes) with fully-connected SNNs, achieving 69% accuracy with Send-on-Delta encoding but without convolutional layers or hardware deployment. Yarga et al. [21] compare four spike encodings

Table 2: ESC-50 accuracy (5-fold CV). All pairwise comparisons are statistically significant ($p < 0.002$) except rate $\approx$ phase ($p=0.93$). Cohen’s d ranges from 3.75 to 8.13.

Model / Encoding	Mean (%)	$\pm$ Std	Significance
ANN baseline	63.85	3.07	—
SNN Direct	47.15	4.50	$p=0.001$ vs ANN
SNN Phase	24.15	1.66	$p < 0.001$ vs direct
SNN Rate	24.00	1.90	$p=0.93$ vs phase
SNN Population	19.15	2.79	$p < 0.002$ vs rate
SNN Latency	16.30	1.62	$p < 0.002$ vs pop.
SNN Delta	7.25	0.94	$p < 0.001$ vs lat.
SNN Burst	6.50	1.54	$p < 0.002$ vs delta

**Figure 3: ESC-50 accuracy across seven SNN encodings and ANN baseline (5-fold CV). Direct encoding dominates all spike-converted methods by >20 pp.**

to distribute synaptic load across multiple ARM cores, preventing DMA buffer overflows when processing high-fan-in layers.

3.4 NeuroBench Energy Analysis

We follow the NeuroBench framework [9] to count effective ACs (SNN) and MACs (ANN). Energy estimation uses Horowitz ISSCC 2014 costs [15]: AC=0.9 pJ, MAC=4.6 pJ.

3.5 Adversarial Robustness Evaluation

We apply FGSM [12] and PGD [13] attacks across seven ϵ values (0.00–0.20) using the torchattacks library. All results are 5-fold validated. Prior work on image classification [11] shows SNNs exhibit inherent adversarial robustness from discrete encoding; we provide the first such analysis on audio spectrograms. We note that Wang et al. [17] recommend SA-PGD for stronger SNN evaluation; our standard PGD results represent a lower bound on SNN robustness.

4 RESULTS

4.1 Encoding Comparison

Direct encoding outperforms all spike-converted alternatives by >20 pp (Table 2, Figure 3). The near-equality of rate (24.00%) and phase (24.15%, $p=0.93$) is notable: phase produces exactly one spike

Table 3: Cross-encoding transfer ratio (5-fold validated). Models trained with one encoding are evaluated using all others. Low ratios indicate high encoding specificity.

Metric	Value
Transfer ratio (mean $\pm$ std)	0.255 ± 0.006
Mean accuracy loss when swapped	75%
Fold-to-fold consistency ($n=5$)	$<2.4\%$ variation

Table 4: SpiNNaker 5-fold results (FC2-only hybrid proof of concept, 2,000 total inferences). Hardware gap is statistically significant ($p=0.0016$, paired t -test).

Fold	SpiNNaker (%)	snnTorch (%)	Gap (pp)
1	29.0	43.25	14.2
2	32.0	44.0	12.0
3	36.5	47.75	11.2
4	43.0	51.25	8.2
5	25.2	43.5	18.3
Mean	33.1 ± 6.9	46.0	12.8 ± 4.1

per neuron while rate produces ~ 7 , yet both achieve statistically identical accuracy. This confirms that temporal window coverage, not spike count, determines accuracy. The 16.7 pp SNN-ANN gap is statistically significant (paired t -test: $p=0.001$, Cohen’s $d=-2.93$).

Why do encodings fail? Delta (7.25%) fails because static spectrograms have no temporal variation. Burst (6.50%) concentrates spikes in the first 5 of 25 timesteps, creating a temporal mismatch that causes severe overfitting (train 45–62%, test 5–9%).

4.2 Encoding Transfer Analysis

We introduce a cross-encoding transfer analysis: models trained with one encoding are evaluated using all other encodings. The transfer ratio (accuracy with mismatched encoding divided by accuracy with matched encoding) averages 0.255 ± 0.006 across all five folds (Table 3). This remarkably consistent result ($<2.4\%$ coefficient of variation) demonstrates that SNN representations are highly encoding-specific—swapping the encoding at test time causes 75% accuracy loss on average. To our knowledge, this is the first quantification of encoding specificity in SNNs on any benchmark.

4.3 SpiNNaker Hardware Results

Table 4 presents the first neuromorphic hardware deployment for environmental sound classification. The 12.8 pp hardware gap ($p=0.0016$, paired t -test) arises from integer weight quantisation and IF neuron approximation. While accuracy is modest, the prior art [8] evaluated only 8 pure tones; our 50-class task is fundamentally harder.

Root-cause analysis of FC₁ failure. Full SpiNNaker deployment requires binary inputs throughout. The AvgPool layer between LIF₂ and FC₁ produces fractional outputs in $[0, 1]$. FC₁ weights have near-zero mean ($\mu=-0.0034$), so 1,398 simultaneous fractional inputs produce net excitatory-inhibitory cancellation ($I_{\text{net}} \approx 0$), yielding zero hidden spikes. Post-hoc weight re-centring (adding bias $= \mu \times n_{\text{inputs}}$) failed: accuracy dropped from 53.75% to 8.50%

Table 5: PANNs CNN14 frozen embeddings + classifier head (5-fold CV). The SNN-ANN gap collapses from 16.7 pp to 0.95 pp.

Head	Mean (%)	$\pm$ Std	vs SNN (p)
PANNs + Linear	93.80	1.69	0.011
PANNs + ANN	93.45	1.54	0.034
PANNs + SNN	92.50	1.30	—

Table 6: Accuracy (%) under adversarial attack (5-fold mean $\pm$ std). Clean accuracy differs from Table 2 due to Batch-Norm running statistics across evaluation hardware; robustness ratios are unaffected (see text).

ϵ	FGSM SNN	FGSM ANN	PGD SNN	PGD ANN
0.00	54.25 $\pm$ 5.1	61.85 $\pm$ 4.6	54.25 $\pm$ 5.1	61.85 $\pm$ 4.6
0.01	33.55 $\pm$ 3.7	21.05 $\pm$ 4.4	18.65	12.65
0.02	29.30 $\pm$ 4.3	9.75 $\pm$ 2.2	13.65	2.05
0.05	22.75 $\pm$ 4.1	5.05 $\pm$ 1.9	9.75	0.05
0.10	16.55$\pm$5.5	2.75$\pm$0.6	3.50	0.00
0.20	13.25 $\pm$ 6.8	2.50 $\pm$ 0.7	1.15	0.00

because the bias compensation assumes binary inputs but actual inputs are fractional. This reveals that standard conv $\rightarrow$ pool $\rightarrow$ FC architectures are not directly deployable on spike-only hardware without architectural redesign—a practically important co-design constraint.

4.4 Transfer Learning: Gap Collapse

The SNN-ANN gap collapses from 16.7 pp (scratch) to 0.95 pp ($p=0.034$) with frozen AudioSet-pretrained CNN14 embeddings [10] (Table 5). Notably, the linear probe (93.80%) outperforms both learned heads, indicating that CNN14 embeddings are already linearly separable for ESC-50. The SNN head ($p=0.011$ vs linear) adds computational overhead without accuracy benefit. Nevertheless, the key scientific insight holds: given equal-quality features, the SNN achieves accuracy within 0.95 pp of the ANN, demonstrating that the scratch-training gap is a *feature-learning* problem, not a *spiking computation* limitation. In a practical hybrid deployment, CNN14 extracts features in software while the SNN head classifies on neuromorphic hardware.

4.5 Adversarial Robustness

Under FGSM at $\epsilon=0.1$, the SNN retains 16.55% $\pm$ 5.49% accuracy versus 2.75% $\pm$ 0.61% for the ANN (**6.0 $\times$ more robust**, $p=0.007$, 5-fold validated; Table 6). Under PGD at $\epsilon=0.05$, the SNN retains 9.75% while the ANN drops to 0.05%. Binary spike thresholding provides natural gradient masking: small perturbations that shift continuous activations have no effect until crossing the LIF threshold.

Clean accuracy note. The $\epsilon=0.00$ values (SNN=54.25%, ANN=61.85%) differ from Table 2 due to BatchNorm running statistics on different evaluation hardware; robustness ratios are unaffected as both numerator and denominator use the same pipeline.

Table 7: NeuroBench energy comparison (5-fold mean $\pm$ std). SNN activation sparsity = 73.6% $\pm$ 0.7% (spike rate = 26.4%).

Model	Eff. Ops	Energy (nJ)	Op Type
SNN	1.08M $\pm$ 41K ACs	968 $\pm$ 37	AC $\times$ 0.9 pJ
ANN	99K $\pm$ 2K MACs	454 $\pm$ 11	MAC $\times$ 4.6 pJ

4.6 Energy Analysis

In software simulation, the ANN is 2.1 $\times$ more energy-efficient (Table 7) because the SNN’s $T=25$ timesteps generate 10.9 $\times$ more operations despite each being cheaper. On neuromorphic hardware, each AC costs 5.1 $\times$ less than each MAC [15]; however, the SNN’s 26.4% spike rate exceeds the <6.4% threshold identified by Dampfhof et al. [6] for energy parity with quantised ANNs. Reducing T and increasing sparsity are necessary for the SNN to achieve energy advantage—this is an important direction for future work.

4.7 Temporal Efficiency

We evaluate the SNN at reduced timestep budgets $T \in \{1, 3, 5, 7, 10, 15, 20, 25\}$ using direct encoding. Accuracy reaches 90% of its full value ($T=25$) at $T=7$, corresponding to a 72% reduction in energy (proportional to T). Performance plateaus around $T=15$ – 20 , with $T=20$ and $T=25$ statistically indistinguishable ($p=0.45$). This suggests that the SNN extracts most task-relevant information within the first 7–10 timesteps, and that aggressive temporal truncation is a viable path to energy efficiency on neuromorphic hardware.

4.8 Continual Learning

We evaluate catastrophic forgetting by sequentially training SNN and ANN on five ESC-50 super-categories (Animals, Nature, Human, Domestic, Urban), each containing 10 classes. After training all five tasks, we measure mean forgetting: the average accuracy drop on previously-learned tasks. Over five folds, the SNN exhibits 69.9% $\pm$ 4.3% forgetting versus 74.7% $\pm$ 2.4% for the ANN—the SNN forgets 4.7 pp less on average. While both models suffer substantial catastrophic forgetting (no replay or regularisation), the SNN’s binary spike representations provide a modest but consistent advantage across 4 of 5 folds. This preliminary result suggests that spiking dynamics may offer a natural inductive bias for sequential learning, warranting further investigation with techniques such as STDP-based consolidation.

4.9 SpiNNaker Per-Category Analysis

Analysis of SpiNNaker super-category performance (fold 4, 400 samples) reveals systematic variation. Natural sounds (61.3%) and Animals (45.0%) outperform Domestic (31.2%) and Urban (31.2%). SpiNNaker achieves near-perfect agreement with snnTorch on some classes (e.g., airplane: SpiNNaker 50.0% vs snnTorch 12.5%, a +37.5 pp advantage) while others collapse entirely (insects, glass_breaking, helicopter: 0%). The overall agreement rate between SpiNNaker and snnTorch predictions is 64.5%, indicating that the hardware gap is not uniform—certain spectral patterns survive quantisation better than others.

Table 8: Surrogate gradient ablation (fold 1, seed 42). Bimodal: 3 functions learn, 5 fail. Spike-rate-escape achieves the best accuracy.

Surrogate	Acc. (%)	Best Epoch	Status
Spike-rate-escape	46.00	50	Learned
Fast sigmoid	44.75	50	Learned
Arctangent	35.75	49	Learned
Straight-through est.	10.25	11	Failed
Sigmoid	2.00	11	Failed
Super-fast-sigmoid	2.00	10	Failed
Triangular	2.75	23	Failed

Table 9: Accuracy (%) under Gaussian noise (5-fold mean $\pm$ std). The SNN degrades directionally less, but the difference is not significant at any SNR.

SNR (dB)	SNN (%)	ANN (%)	p -value
Clean	54.25 $\pm$ 5.1	61.85 $\pm$ 4.6	0.003
40	50.35 $\pm$ 5.0	56.10 $\pm$ 4.8	0.07
20	37.55 $\pm$ 3.8	39.50 $\pm$ 4.2	0.48
10	19.85 $\pm$ 2.6	19.15 $\pm$ 3.3	0.72
5	11.05 $\pm$ 1.8	10.25 $\pm$ 2.1	0.52
0	7.05 $\pm$ 1.3	6.95 $\pm$ 1.4	0.94

4.10 Surrogate Gradient Sensitivity

We evaluate eight surrogate gradient functions on fold 1 (Table 8) to assess sensitivity to this design choice. Results show a bimodal distribution: three functions achieve meaningful learning while five fail catastrophically ($\leq 10.25\%$). This bimodality contrasts with Zenke & Vogels [5], who found surrogate shape matters less than slope for image classification. For audio spectrograms, the surrogate function is a critical hyperparameter; spike-rate-escape—motivated by escape noise theory—achieves +1.25 pp over our default fast sigmoid.

4.11 Noise Robustness

We evaluate SNN and ANN accuracy under additive Gaussian noise at five signal-to-noise ratios (Table 9), 5-fold validated. The SNN degrades directionally less than the ANN (relative accuracy drop: SNN 93.8% vs ANN 95.1% from clean to 0 dB). However, the difference is not statistically significant at any individual SNR level ($p > 0.07$). At 0 dB SNR, both models converge near chance. While binary spike thresholding provides some natural noise filtering, this effect is insufficient to produce a statistically significant advantage for our architecture and dataset size.

5 DISCUSSION

Why does direct encoding dominate? Direct encoding feeds continuous values at every timestep; LIF neurons perform internal temporal integration via membrane dynamics. Spike-converted inputs lose information through discretisation: rate discards magnitude, latency collapses temporal richness, and delta amplifies noise

on static inputs. Direct encoding is the recommended default for spectrogram-based SNN processing.

The gap is feature-learning, not spiking. The PANNs result (Table 5) confirms: given equal-quality features, $\text{SNN} \approx \text{ANN}$. The 16.7 pp scratch-training gap reflects difficulty learning convolutional filters from 1,600 samples with approximate surrogate gradients [4] and binary spike bottlenecks, consistent with Deng & Gu [16]. The linear probe (93.80%) outperforming both heads confirms CNN14 features are already well-separated.

Encoding specificity as a design constraint. The transfer ratio of 0.255 means SNNs retain only 25.5% accuracy with mismatched encodings. This high specificity—the SNN learns encoding-dependent representations—has practical implications for neuromorphic systems where the sensor encoding may be fixed by hardware.

SpiNNaker: proof of concept with honest limitations. Our FC2-only hybrid demonstrates feasibility but is not full neuromorphic inference, following the hybrid paradigm explored by Seekings et al. [22]. The FC₁ cancellation (Section 4.3) reveals that standard conv $\rightarrow$ pool $\rightarrow$ FC architectures with AvgPool are not directly deployable on spike-only hardware. Replacing AvgPool with MaxPool (which preserves binary outputs) could enable full deployment on SpiNNaker 2 [18], as demonstrated by Arfa et al. [23] for gesture recognition.

Adversarial robustness. Binary thresholding provides natural gradient masking, consistent with Sharmin et al. [11] on images. We acknowledge that standard PGD may not be optimal for evaluating SNN robustness [17]; SA-PGD evaluation is left for future work. Our FGSM results (single-step, no SNN-specific adaptation) provide a reliable lower bound.

Temporal truncation as energy strategy. Our temporal ablation shows the SNN reaches 90% accuracy at $T=7$, providing a concrete path to energy efficiency: at $T=7$, the SNN would require only $0.28 \times 1.08\text{M} = 302\text{K}$ effective ACs (272 nJ), making it 1.7 $\times$ cheaper than the ANN in software and dramatically cheaper on neuromorphic hardware. This approaches the <6.4% spike rate threshold identified by Dampfhofer et al. [6].

Continual learning implications. The 4.7 pp forgetting reduction (SNN 69.9% vs ANN 74.7%) is modest but consistent across 4/5 folds. We hypothesise this arises from the SNN’s sparse, binary activations: each task activates a distinct subset of neurons, providing a natural form of representation separation. However, neither model uses any continual learning mechanism—the observed advantage is purely from architectural inductive bias. Combining SNNs with STDP-based consolidation [20] could amplify this effect.

Surrogate gradient choice matters for audio. The bimodal distribution of surrogate function performance (3 functions learn, 5 fail) is a practical finding: for audio spectrogram classification, researchers should use spike-rate-escape, fast sigmoid, or arctangent, and avoid the straight-through estimator or sigmoid surrogates. This contrasts with image classification where surrogate choice is reportedly less critical [5].

Threats to validity. (1) Results are for a single architecture (~622K params); wider/deeper networks may shift the encoding hierarchy. (2) ESC-50 is small (2,000 clips); generalisation to UrbanSound8K or FSD50K is untested. (3) NeuroBench energy is computed in software simulation; actual SpiNNaker power consumption was not measured. (4) The surrogate gradient ablation uses a single fold and seed; full 5-fold validation may change the ranking. (5) Noise robustness and continual learning results, while 5-fold validated, are directional trends that do not reach statistical significance at individual test points.

6 CONCLUSION

We present the first convolutional SNN evaluation on ESC-50, establishing a 47.15% baseline with direct encoding and deploying on SpiNNaker neuromorphic hardware as a proof of concept. The key finding is that the SNN-ANN gap is not fundamental—it collapses from 16.7 pp to 0.95 pp with pre-trained features ($p=0.034$), demonstrating that the bottleneck is feature learning, not spiking computation.

Additional contributions include: $6.0\times$ greater adversarial robustness under FGSM ($p=0.007$, 5-fold validated), a novel cross-encoding transfer analysis revealing high encoding specificity (transfer ratio = 0.255 ± 0.006), temporal efficiency analysis showing 90% accuracy at $T=7$ (72% energy saving), and continual learning analysis demonstrating 4.7 pp less forgetting for SNNs (5-fold validated). These results collectively characterise the strengths and limitations of SNNs for environmental sound classification.

Future work includes SpiNNaker 2 [18] deployment with binary-compatible architectures (replacing AvgPool with MaxPool to enable full FC_1+FC_2 hardware inference), sparsity optimisation to reach the $<6.4\%$ spike-rate threshold for neuromorphic energy parity, STDP-based continual learning, and cross-dataset validation on UrbanSound8K and FSD50K.

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